RAG Learning

Required Practicals (RP1–RP10)

AQA GCSE Physics (8463) — separate Physics

RP1 — Specific Heat Capacity

Aim: Determine the specific heat capacity of a metal block or liquid (e.g. water) using an electrical heater. Specific heat capacity (c) is the energy required to raise the temperature of 1 kg of a substance by 1 °C.

Apparatus

- Metal block (or insulated beaker for liquid) with two holes: one for immersion heater, one for thermometer
- 12 V immersion heater (electric heater)
- Joulemeter (or ammeter + voltmeter + stopwatch)
- Thermometer (0–110 °C)
- Lagging (thermal insulation, e.g. cotton wool, fibreglass, or foam)
- Balance (±0.01 g)
- Power supply (12 V AC or DC)

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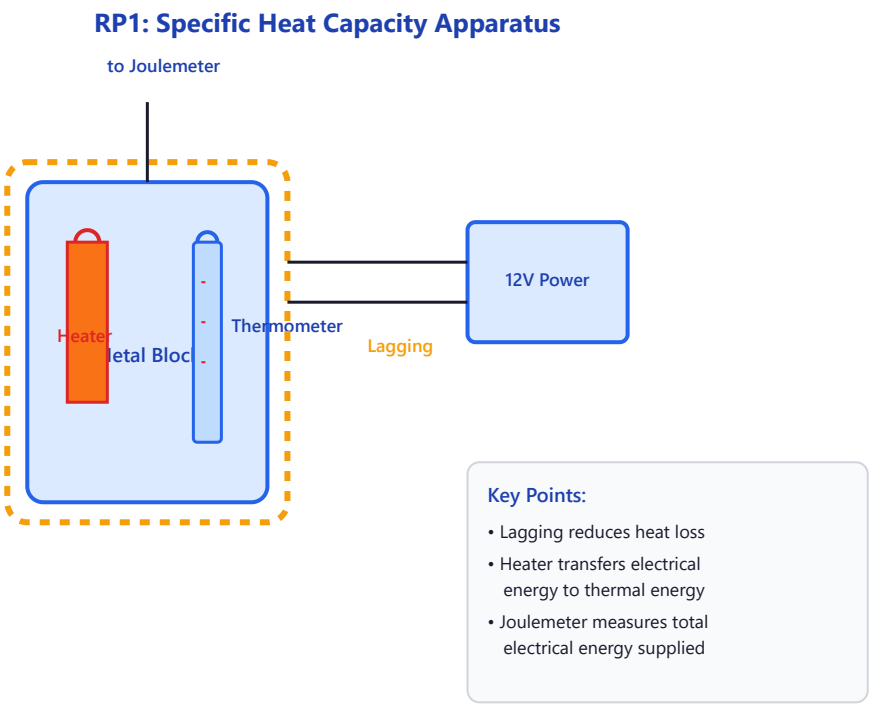


Method

1. Measure and record the **mass** of the metal block (or liquid in beaker) using the balance.
2. Record the **initial temperature** θ_1 (thermometer reading).
3. Switch on the power supply, starting the heater and noting the **initial reading on the joulemeter** (or start the stopwatch and note ammeter/voltmeter readings).
4. Allow the block to heat up, recording the thermometer reading at regular intervals (e.g. every 2–3 minutes).
5. When the temperature has risen by a suitable amount (e.g. $\Delta\theta$ = 10–20 °C), record the **final temperature** θ_2 and the **final joulemeter reading** (or stop the stopwatch and note the final meter readings).
6. Switch off the power supply and allow the block to cool.
7. Calculate the energy supplied by the heater:
   `E = Joulemeter reading (in joules)`
   or `E = I × V × t` (amperes × volts × seconds)
8. Calculate the temperature rise: $\Delta\theta = \theta_2 - \theta_1$
9. Use the formula: $c = E / (m \times \Delta\theta)$ to find specific heat capacity.

Key Measurements and Calculations

- **Mass (m):** in kg (convert from grams if necessary)
- **Initial and final temperatures:** in °C (or K, since Δθ is the same in both)
- **Energy supplied (E):** in joules (J)
- **Specific heat capacity formula:** `c = E / (m × Δθ)`
- **Units:** J/(kg·°C) or J/(kg·K)
- **Typical values:** water ~4200 J/(kg·K); aluminum ~900 J/(kg·K); iron ~450 J/(kg·K)

Sources of Error & Minimisation

- **Heat loss to surroundings:** Use lagging (thermal insulation) around the block to reduce cooling. Measure the final temperature a minute after switching off to account for any heat still being transferred.
- **Thermometer calibration error:** Use a thermometer with small scale divisions (0.5 °C or finer). Record readings at eye level to avoid parallax error.
- **Uneven heating of block:** Allow time for temperature equilibrium within the block; stir liquid samples if used.
- **Electrical measurement error:** Ensure ammeter and voltmeter are correctly calibrated. Use a joulemeter for direct energy measurement if available (more accurate than V × I × t).

Safety Considerations

- **Hot apparatus:** The block and heater will reach high temperatures (>80 °C). Use insulated tongs or gloves when handling; allow to cool before moving.
- **Electrical safety:** Check the power lead for damage; do not use if wet. Keep water away from electrical connections.
- **Thermometer:** If glass, handle carefully to avoid breakage and mercury spillage (if mercury thermometer used).

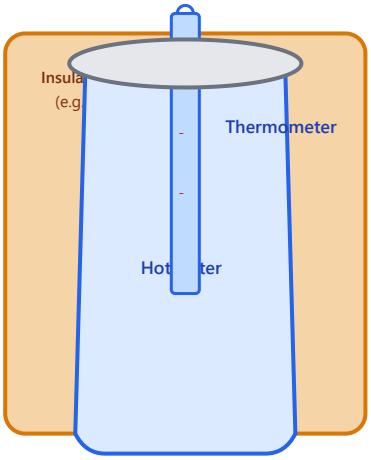
RP2 — Thermal Insulation

Aim: Investigate which material is the best thermal insulator by comparing the cooling curves of containers wrapped in different insulating materials.

Apparatus

- Glass beaker or metal can (e.g. 500 mL)
- Lid (with hole for thermometer)
- Insulating materials: bubble wrap, cotton wool, felt, newspaper, foam, etc.
- Thermometer (0–110 °C)
- Measuring cylinder (to measure volume of hot water)
- Stopwatch
- Access to boiling water (kettle or Bunsen burner)
- Data logger with temperature sensor (optional, for continuous recording)

RP2: Thermal Insulation Apparatus (Cross-section View)



Measurements:

t = 0 min: θ = 70 °C
t = 1 min: θ = 68 °C
t = 2 min: θ = 66 °C
... (every 1 min)
t = 10 min: θ = 45 °C

Test different materials:

Bubble wrap | Cotton wool | Felt | Newspaper | Foam | Polystyrene chips

Method

1. Measure a fixed volume of boiling water (e.g. 300 mL) using a measuring cylinder.
2. Pour the hot water into the beaker and immediately place the thermometer through the lid.

3. Record the **initial temperature** (θ_0) and the **time** (t = 0).
4. Wrap the beaker completely with the first insulating material (e.g. bubble wrap, 2–3 cm thickness).
5. Record the temperature at regular intervals (e.g. every 1 minute) for 10 minutes.
6. Allow the beaker to cool to room temperature.
7. Repeat steps 1–6 with each of the other insulating materials, ensuring the same initial temperature and volume of water each time.
8. Plot cooling curves (temperature vs. time) for each material on the same graph.

Key Measurements and Calculations

- **Volume of water:** must be the same for all tests (e.g. 300 mL)
- **Initial temperature:** should be approximately the same (~70 °C) for each test
- **Temperature readings:** recorded at fixed time intervals
- **Cooling rate:** gradient of cooling curve = $\Delta T / \Delta t$ (°C/min)
- **Best insulator:** lowest cooling rate (flattest gradient) or highest final temperature after 10 minutes

Sources of Error & Minimisation

- **Variation in initial temperature:** Use boiling water each time (though it cools slightly in the journey from kettle to beaker). Record the initial reading immediately.
- **Variation in room temperature:** Conduct all tests in the same location at the same time of day if possible.
- **Variation in water volume:** Use a measuring cylinder to ensure each test uses exactly the same volume (e.g. 300 mL).
- **Inconsistent insulation thickness:** Wrap each material to the same thickness (e.g. measure with ruler).
- **Temperature measurement error:** Read thermometer at eye level; use data logger for continuous, precise readings if available.

Safety Considerations

- **Boiling water:** Wear appropriate eye protection and heat-resistant gloves. Be careful when pouring hot water.
- **Hot apparatus:** Do not touch the beaker or insulation for several minutes after setup.

RP3 — Resistance of a Wire

> **Aim:** Investigate how electrical resistance varies with the length of a wire (at constant cross-sectional area and material). Expect: Resistance ∝ Length.

Apparatus

- Resistance wire (constantan or nichrome, e.g. 0.5 mm diameter)
- Metre ruler (calibrated to mm)
- Crocodile clips (two)
- Ammeter (0–1 A or 0–2 A, digital or analogue)
- Voltmeter (0–5 V, digital or analogue)
- 6 V power supply (low voltage for safety)
- Switch
- Leads and connecting wire

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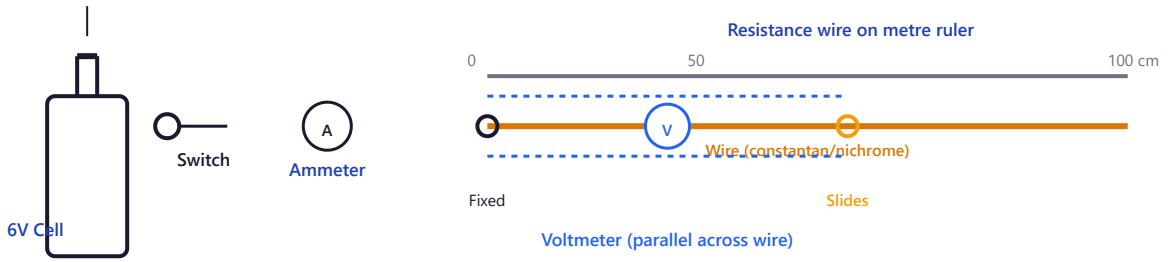
RP3: Resistance of a Wire — Circuit Diagram



Procedure:

1. Set crocodile clip at length L = 10 cm. 2. Close switch briefly. 3. Record ammeter (I) and voltmeter (V) readings. 4. Calculate R = V/I. 5. Repeat at L = 20, 30, ..., 1(

Method

1. Set up the circuit: 6V cell → Switch → Ammeter → Resistance wire (between crocodile clips) → back to cell.
2. Connect the voltmeter in parallel across the resistance wire (between the two crocodile clips).
3. Tape the resistance wire to the metre ruler, with one crocodile clip at the 0 cm mark (fixed).
4. Position the sliding crocodile clip at **L = 10 cm**.
5. Close the switch briefly and record the ammeter reading (I) and voltmeter reading (V). Open the switch immediately to avoid overheating the wire.
6. Calculate resistance: `R = V / I` (in ohms, Ω).
7. Move the sliding clip to **L = 20 cm** and repeat steps 5–6.
8. Continue at L = 30, 40, 50, 60, 70, 80, 90, 100 cm.
9. Tabulate the results (L, I, V, R).
10. Plot a graph of Resistance (y-axis) vs. Length (x-axis). Expect a straight line through (or very close to) the origin.

Key Measurements and Calculations

- **Length (L):** distance between the two crocodile clips, in cm or m

- **Current (I):** ammeter reading, in amperes (A)
- **Voltage (V):** voltmeter reading, in volts (V)
- **Resistance (R):** $R = V / I$ (ohms, Ω)
- **Relationship:** $R \propto L$ (resistance is directly proportional to length)
- **Gradient of graph:** ρ / A (resistivity divided by cross-sectional area)

Sources of Error & Minimisation

- **Wire heating during measurement:** Close the switch for only a few seconds to take readings; do not leave it on. Use low currents (<0.5 A) to minimize heating. Allow wire to cool between readings.
- **Non-uniform wire cross-section:** Use a high-quality resistance wire with consistent diameter. If diameter varies, results will scatter.
- **Meter calibration error:** Check that meters read zero when not in use. Use calibrated digital meters if available for greater accuracy.
- **Poor electrical contact at crocodile clips:** Ensure clips bite firmly into the wire; clean wire with emery paper if needed.

Safety Considerations

- **Wire heating:** The wire may become hot if left on for prolonged periods. Do not touch without allowing time to cool.
- **Low voltage used:** 6 V is safe, but avoid touching the circuit with wet hands.

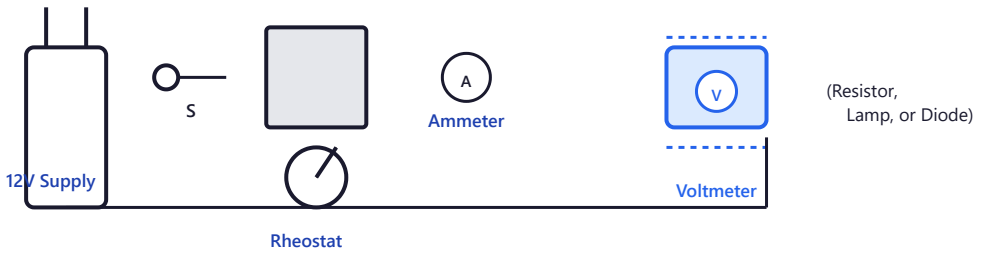
RP4 — I-V Characteristics

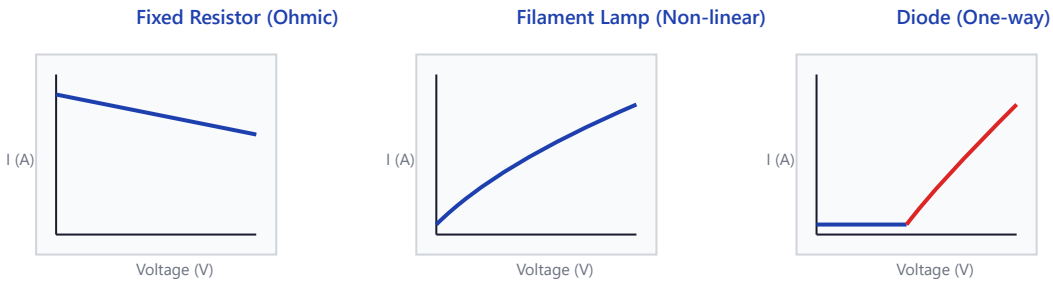
> **Aim:** Investigate the current-voltage (I-V) characteristics of a fixed resistor, filament lamp, and diode by varying voltage and measuring current. Components exhibit different I-V relationships: ohmic, non-linear, and one-way conduction.

Apparatus

- Variable resistor (rheostat, e.g. 10 Ω or 50 Ω)
- Ammeter (0–2 A or 0–5 A)
- Voltmeter (0–10 V)
- Power supply (12 V AC or DC, or dual voltage supply)
- Switch
- Test components: fixed resistor (e.g. 10 Ω), filament lamp (e.g. 12 V 0.5 A), diode (e.g. 1N4007)
- Connecting leads and wires

RP4: I-V Characteristics — Circuit Diagram





Method

1. Set up the circuit: 12V supply → Switch → Rheostat (variable resistor) → Ammeter → Test component → back to supply. Connect Voltmeter in parallel across the test component.
2. Begin with the test component being a **fixed resistor**.
3. Adjust the rheostat to the maximum resistance (minimum current). Close the switch.
4. Gradually decrease the rheostat resistance, taking ammeter and voltmeter readings at 10–12 different voltage levels (e.g. 0 V, 1 V, 2 V, … 10 V).
5. Record all (V, I) pairs in a table.

6. Reverse the polarity of the power supply (swap the leads) and repeat the measurement, recording the negative voltage and current values.
7. Plot a graph of I (y-axis) vs. V (x-axis) for the fixed resistor. Expect a straight line through the origin (ohmic behavior).
8. Repeat steps 2–7 with the **filament lamp**. Expect an S-shaped or curved line (non-ohmic; resistance increases with temperature and current).
9. Repeat steps 2–7 with the **diode**. Expect zero current in reverse bias; sharp rise in current when forward-biased (typically above ~0.6–0.7 V for silicon diodes).

Key Measurements and Calculations

- **Voltage (V):** voltmeter readings, in volts
- **Current (I):** ammeter readings, in amperes
- **Resistance (R):** R = V / I for any data point
- **Fixed resistor:** R is constant (ohmic); graph is a straight line through origin.
- **Filament lamp:** R increases with temperature (and current); graph curves upward (non-linear).
- **Diode:** R is very high in reverse bias (≈∞); R drops sharply once forward-biased (≈0.6–0.7 V for Si).

Sources of Error & Minimisation

- **Component heating:** For filament lamp and diode, heating can change resistance. Take measurements quickly and allow cooling time between readings.
- **Meter calibration error:** Use calibrated digital meters. Check that meters read zero when off.
- **Temperature variation in filament lamp:** Record measurements in the same order (forward and reverse) each time. Note that filament resistance varies significantly with temperature.
- **Polarity confusion with diode:** Mark the diode legs clearly (anode vs. cathode) to avoid repeated reversals.

Safety Considerations

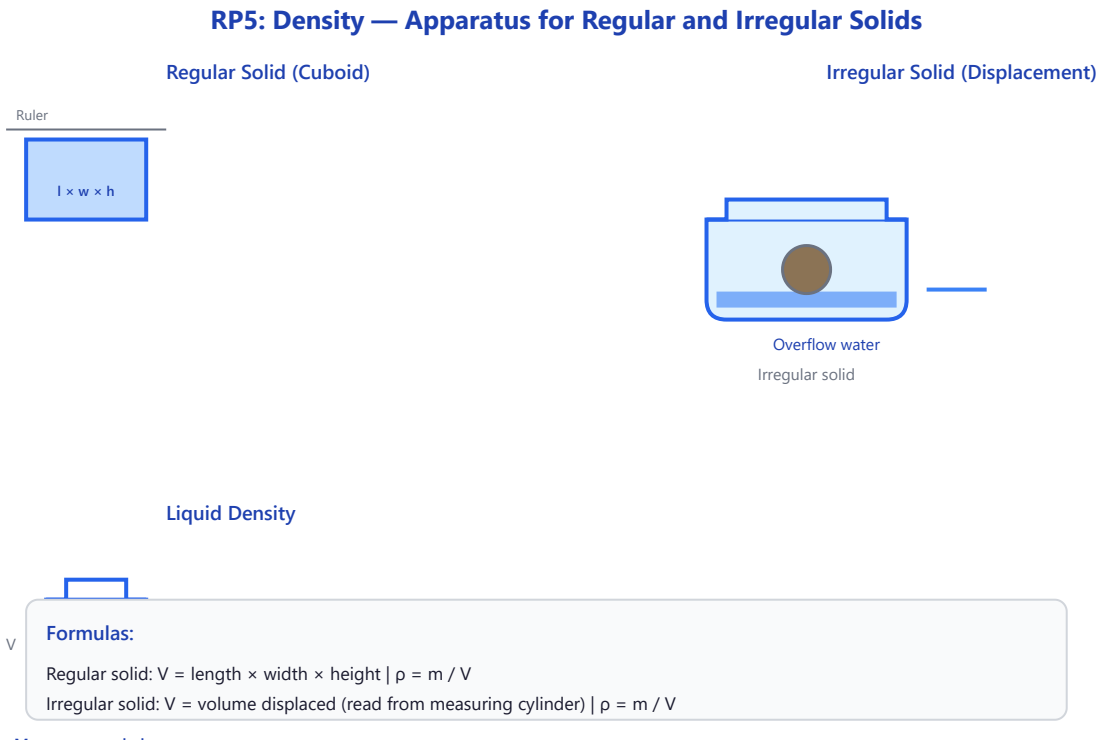
- **Filament lamp heating:** The bulb becomes very hot; do not touch. Allow to cool before handling.
- **12 V supply:** Generally safe, but avoid touching live connections with wet hands.

RP5 — Density

Aim: Determine the density of regular solids, irregular solids, and liquids using mass and volume measurements. Density (ρ) = mass / volume.

Apparatus

- Balance (±0.01 g or better)
- Ruler or vernier caliper (for measuring regular solid dimensions)
- Measuring cylinder (e.g. 100 mL or 250 mL, graduated to 1 mL)
- Displacement can (Eureka can) or overflow can (for irregular solids)
- Beaker (for collecting displaced water)
- Regular solids to measure: metal cuboid, cylinder, etc.
- Irregular solids to measure: rock, metal piece, etc.
- Liquids to measure: water, oil, salt solution, etc.



Method — Regular Solids

1. Measure the **mass** of the solid using the balance. Record to nearest 0.01 g.
2. Using a ruler or vernier caliper, measure the **dimensions** of the solid:
   - **Cuboid:** length, width, height (in cm or m)
   - **Cylinder:** diameter (or radius) and height
3. Calculate the **volume**:
   `V = l × w × h` (cuboid, in cm^3)
   `V = π × r² × h` (cylinder, in cm^3)
4. Calculate the **density**:
   `ρ = m / V` (g/cm^3 or kg/m^3)

Method — Irregular Solids

1. Measure the **mass** of the solid using the balance.
2. Fill a displacement can (Eureka can) with water until it flows out of the spout into a beaker.
3. When water stops flowing, carefully lower the irregular solid into the can using a string or holder.
4. Collect all the water that overflows into the beaker.
5. Pour the displaced water into a measuring cylinder and record the **volume** (in mL or cm^3).
6. Calculate the **density**:
   `ρ = m / V`

Method — Liquids

1. Measure the mass of an **empty measuring cylinder** using the balance. Record as m_1.
2. Pour the liquid into the measuring cylinder to a convenient mark (e.g. 50 mL or 100 mL) and record the **volume** (V) at eye level (meniscus at bottom).
3. Measure the mass of the cylinder **with liquid**. Record as m_2.
4. Calculate the mass of the liquid: $m = m_2 - m_1$
5. Calculate the **density**:
   `ρ = m / V`

Key Measurements and Calculations

- **Mass (m):** in grams (g) or kilograms (kg)

- **Volume (V):** in cm^3 (or mL) or m^3
- **Density (ρ):** $\rho = m / V$, in g/cm^3 or kg/m^3
- **Typical values:** water ~1.0 g/cm^3; aluminum ~2.7 g/cm^3; iron ~7.9 g/cm^3; lead ~11.3 g/cm^3
- **Conversion:** 1 g/cm^3 = 1000 kg/m^3

Sources of Error & Minimisation

- **Parallax error (reading measuring cylinder):** Always read the meniscus at eye level. For water, read from the bottom of the meniscus curve.
- **Incomplete submersion (irregular solids):** Ensure the solid is fully submerged in the displacement can; if it floats, use a holder or weight to keep it under.
- **Water spillage in displacement method:** Use a tray to catch spilled water. Ensure the overflow spout is at the correct height before the solid is inserted.
- **Air bubbles on solid surface:** Pat the solid dry with tissue before weighing; ensure no air bubbles are attached when submerging.
- **Balance error:** Check that the balance reads zero when empty. Use a consistent balance for all measurements.

Safety Considerations

- **Water spillage:** Keep area dry to avoid slipping.
- **Heavy solids:** Use care when handling dense objects (lead, iron); avoid dropping on feet.

RP6 — Force and Extension (Hooke's Law)

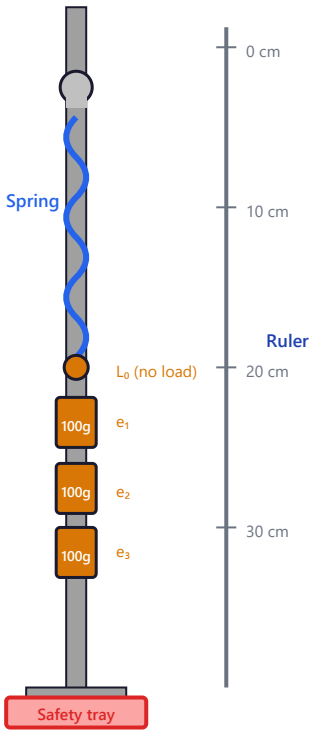
> **Aim:** Investigate the extension of a spring as force is increased, demonstrating Hooke's law: $F = ke$, where F is force, k is the spring constant, and e is extension.

Apparatus

- Spring (low stiffness, e.g. 100 g/cm or 200 g/cm)
- Retort stand, boss, and clamp (to hang spring vertically)

- Metre ruler (or 30 cm ruler, fixed vertically beside spring)
- Mass hanger (e.g. 50 g)
- Slotted masses (100 g each, or 50 g increments)
- Pointer (optional, card strip to help with ruler reading)
- Safety tray (to catch falling masses)

RP6: Hooke's Law — Apparatus



Procedure:

1. Set up stand, clamp spring vertically.
2. Attach hanger (unloaded); note position on ruler (L_0).
3. Add 100 g mass; record new length L_1.

Calculate: $e = L - L_0$

Calculate: $F = mg$ (in N)

4. Repeat with 200 g, 300 g, 400 g, 500 g.
5. Plot F vs. e graph.

Expect: straight line through origin (within elastic limit)

Gradient = k (spring constant)

Beyond limit: curve bends (permanent deformation)

Method

1. Set up the retort stand with the clamp at the top. Hang the spring from the clamp.
2. Position a vertical metre ruler beside the spring so you can read the position easily.
3. Attach the mass hanger (empty) to the spring's lower end. Record the **unstretched length** L_0 (or initial pointer position on the ruler).
4. Add a **100 g mass** to the hanger. Wait for equilibrium and record the new length L_1.
5. Calculate the extension: `e = L − L₀` (in cm)
6. Calculate the force: `F = mg = 0.1 × 10 = 1 N` (use $g = 10\ m/s^2$ for simplicity)

7. Repeat steps 4–6 by adding more masses: 200 g, 300 g, 400 g, 500 g (or until spring is noticeably stretched).
8. Tabulate all results (mass in g, F in N, L in cm, e in cm).
9. Plot a graph of **Force (N) vs. Extension (cm)**. Use the first few data points (within elastic limit).
10. Measure the gradient of the straight section: gradient = k (spring constant, in N/cm or N/m).
11. Verify Hooke's law: `F = ke`

Key Measurements and Calculations

- **Unstretched length (L_0):** initial position on ruler, in cm
- **Loaded length (L):** position on ruler with masses, in cm
- **Extension (e):** $e = L - L_0$, in cm or m
- **Force (F):** F = mg, in newtons (N). Use g = 10 m/s^2 or 9.8 m/s^2
- **Spring constant (k):** k = F / e (N/cm or N/m)
- **Hooke's law:** F = ke (within elastic limit)

Sources of Error & Minimisation

- **Parallax error (reading ruler):** Always read the ruler at eye level, with the pointer aligned with the ruler mark.
- **Exceeding elastic limit:** Spring may not return to original length if stretched too far. Add masses carefully and stop if permanent deformation occurs (spring doesn't return when unloaded).
- **Spring vibration:** Allow spring to settle before recording each reading (wait ~10 seconds after adding a mass).
- **Friction in clamp:** Ensure clamp is tight enough to prevent slipping but not so tight that it damages the spring.
- **Initial loading:** The mass hanger itself stretches the spring slightly; account for this by measuring L_0 with the hanger already attached (no additional masses).

Safety Considerations

- **Falling masses:** Use a safety tray beneath the apparatus. Do not work below the spring when masses are loaded.

- **Eye protection:** Wear safety glasses in case a mass slips and falls.
- **Spring breakage:** Do not exceed the elastic limit; do not exceed the safe loading given by the manufacturer.

RP7 — Acceleration

> **Aim:** Investigate how acceleration depends on (a) the resultant force for a fixed total mass, and (b) the mass for a fixed resultant force. Demonstrates Newton's second law: F = ma.

Apparatus

- Trolley (low-friction, ~500 g)
- Air track or friction-compensated runway (low-friction surface)
- Light gates (two) or ticker timer with paper tape
- Data logger (for light gates)
- String and pulley
- Slotted masses (e.g. 100 g each)
- Mass hanger (e.g. 10 g)
- Ruler (for measuring distances between light gates or tape marks)

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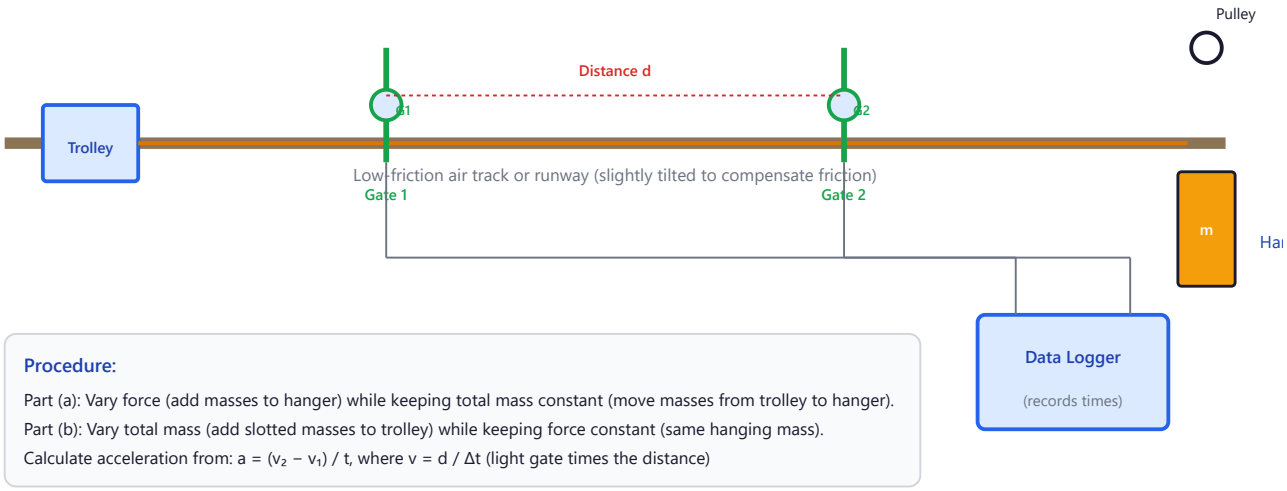
RP7: Acceleration — Apparatus (Light Gates Method)



Method

1. Set up the air track or runway slightly tilted to compensate for friction (nearly horizontal).
2. Place the trolley on the track and attach a string that runs over a pulley to a hanging mass hanger.
3. Position two light gates along the track at a known distance apart (e.g. 50 cm). Connect to a data logger to measure transit times.
4. Start with a fixed setup:
   - **Trolley mass (m_t):** trolley + some slotted masses
   - **Hanging mass (m_h):** small mass on hanger (e.g. 10 g)
   - **Total system mass:** m_total = m_t + m_h
5. **Part (a) — Vary force, constant total mass:**
   - Release the trolley and record the time taken to pass through each gate (Δt_1 and Δt_2).
   - Calculate velocities: $v_1 = d / \Delta t_1$ and $v_2 = d / \Delta t_2$ (where d is the gate separation).
   - Calculate acceleration: $a = (v_2 - v_1) / \Delta t_between$ (or use data logger analysis).
   - Add another mass to the hanger (move one from the trolley to the hanger, keeping total constant).
   - Repeat the measurement. Record new force F = m_h × g and acceleration a.
   - Repeat for 3–4 different hanging masses (and corresponding trolley mass reductions).
   - Plot a graph of **a vs. F**; expect a straight line through origin ($a \propto F$).

6. **Part (b) — Vary mass, constant force:**
   - Return to original setup. Keep hanging mass constant (e.g. 10 g).
   - Measure acceleration.
   - Add a slotted mass (100 g) to the trolley, keeping hanging mass the same.
   - Repeat measurements for 4–5 different trolley masses.
   - Plot a graph of **a vs. 1/m_total** (or a vs. m_total); expect a straight line or inverse relationship ($a \propto 1/m$).

Key Measurements and Calculations

- **Distance between gates (d):** in metres, e.g. 0.5 m
- **Transit times:** Δt_1 and Δt_2 (through each gate), in seconds
- **Velocities:** $v = d / \Delta t$ (m/s)
- **Acceleration:** $a = (v_2 - v_1) / t_between$ (m/s^2)
- **Force:** $F = m_h \times g$ (newtons, where m_h is in kg)
- **Newton's second law:** $F = m_total \times a$
- **Part (a) gradient:** F vs. a should give k = m_total (mass, in kg)
- **Part (b) gradient:** a vs. F should be constant; 1/a vs. m_total should be linear

Sources of Error & Minimisation

- **Friction on track:** Tilt the track slightly to compensate for friction. The trolley should move at constant velocity when released without the hanging mass pulling it.
- **Air resistance:** Negligible at low speeds but becomes noticeable at high speeds. Keep experiments at reasonable speeds.
- **Light gate inaccuracy:** Ensure gates are aligned perpendicular to the trolley's path. Use identical gates for both measurements.
- **Timing errors:** Use a data logger for automatic timing rather than manual stopwatch timing (much more accurate).
- **String slipping on pulley:** Ensure pulley is smooth and string is taut throughout the experiment.

Safety Considerations

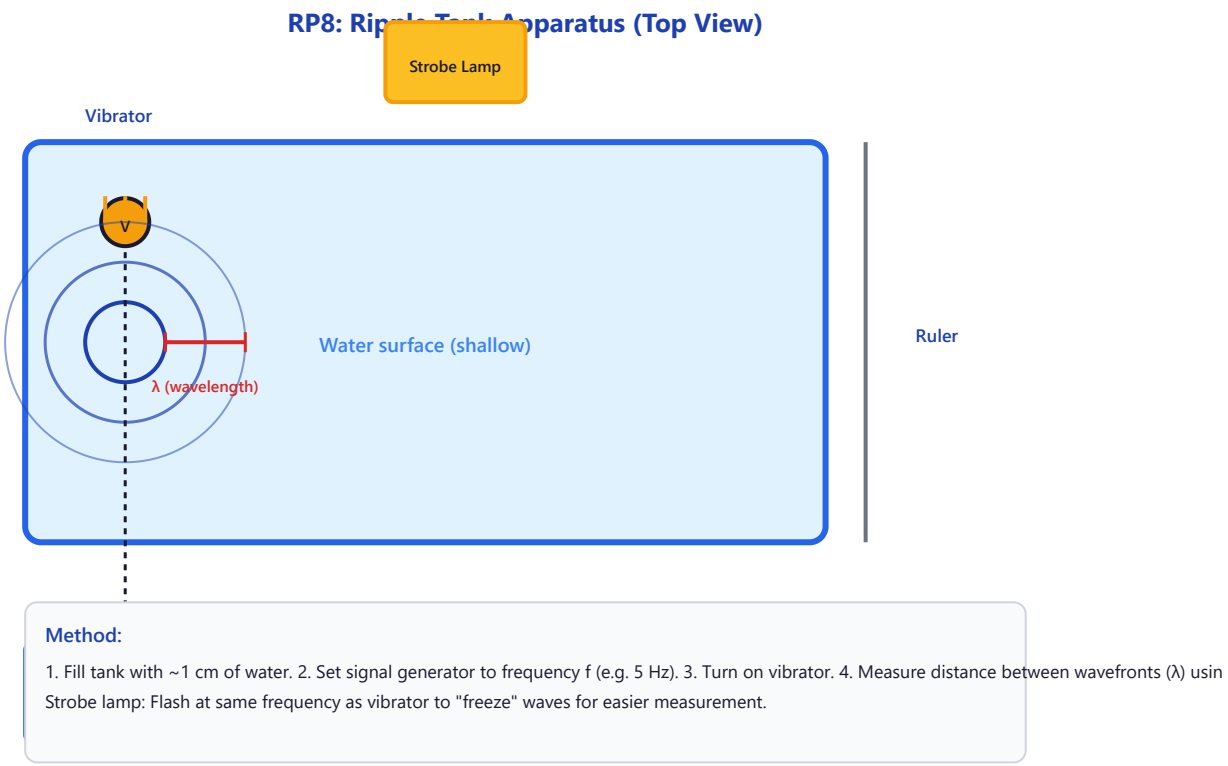
- **Trolley motion:** Keep fingers and loose clothing clear of the moving trolley and pulley.
- **Falling masses:** Ensure the hanging mass falls into a container (beaker) at the end of the track.

RP8 — Waves (Ripple Tank or String)

Aim: Measure the frequency, wavelength, and speed of waves using a ripple tank (water waves) or oscillator on a stretched string. Wave speed $v = f \times \lambda$.

Apparatus — Ripple Tank Method

- Ripple tank (clear plastic or glass tank with shallow water)
- Vibrator (mechanical or electromagnetic) attached to signal generator
- Signal generator (1–50 Hz range typical)
- Strobe lamp or frequency-matched illumination (optional, for freezing waves)
- Screen or white card beneath tank (for shadow visualization)
- Ruler (for measuring wavelength)
- Water (distilled or tap water)



Method — Ripple Tank

1. Fill the ripple tank with water to a shallow depth (approximately 1–2 cm).
2. Attach the vibrator to the signal generator. Set the frequency to a starting value (e.g. 5 Hz).
3. Turn on the vibrator and observe waves spreading across the tank surface.
4. Using a ruler placed across the tank (or a strobe lamp synchronized to the frequency), measure the distance between successive wavefronts (the **wavelength λ**).
5. Record the frequency f from the signal generator.
6. Calculate the wave speed: `v = f × λ`
7. Vary the frequency (e.g. 5, 8, 10, 12 Hz) and repeat measurements to show how wavelength changes while wave speed remains approximately constant (in the same medium).

Apparatus — Stretched String Method

- String (length ~1–2 m, low tension)
- Oscillator (vibrator attached to signal generator, ~10–50 Hz)
- Fixed support (retort stand or clamp)
- String support/fixed end

- Ruler (for measuring wavelength)
- Signal generator

Method — Stretched String

1. Set up the string: one end attached to the oscillator, the other end fixed to a support (or held lightly to allow reflection).
2. Adjust the tension in the string (by raising/lowering the fixed support) until standing waves form at a convenient frequency (e.g. 20–30 Hz).
3. Observe **standing wave patterns** (nodes and antinodes) along the string's length.
4. Count the number of half-wavelengths along the string. If there are n half-wavelengths, then the full wavelength $\lambda = 2 \times (\text{string length}) / n$.
5. Alternatively, measure the distance between adjacent nodes (or antinodes) directly; this distance $= \lambda / 2$.
6. Record the frequency f from the signal generator.
7. Calculate the wave speed: `v = f × λ`
8. Vary the frequency and observe how the standing wave pattern changes (more nodes at higher frequency).

Key Measurements and Calculations

- **Frequency (f):** from signal generator, in hertz (Hz)
- **Wavelength (λ):** distance between successive wavefronts (ripples) or twice the distance between nodes, in metres (m) or centimetres (cm)
- **Wave speed (v):** $v = f \times \lambda$ (m/s)
- **Period (T):** $T = 1 / f$ (seconds)
- **Expected behaviour:** In a ripple tank with fixed water depth, v remains constant; as f increases, λ decreases proportionally.

Sources of Error & Minimisation

- **Measuring wavelength (ripple tank):** Water movements and reflections from tank edges can distort the wave pattern. Use a strobe lamp synchronized to the vibrator frequency to "freeze" the

waves, making measurement easier and more accurate.

- **Vibrator alignment (ripple tank):** Ensure the vibrator just touches the water surface without splashing or submerging. Adjust height if needed.
- **Tank reflections:** Waves reflect from the tank edges and create interference patterns. Measure wavelength away from edges and early in the experiment before reflections become prominent.
- **String tension (oscillator method):** Inconsistent tension changes wave speed. Keep tension fixed during measurements.
- **Frequency calibration:** Check the signal generator against an independent frequency counter if high accuracy is needed.

Safety Considerations

- **Electrical safety (signal generator):** Use a low-voltage supply. Avoid contact with water and electrical connections simultaneously.
- **Strobe lamp (if used):** Flashing lights can induce discomfort. Keep exposure time short and take breaks between observations.

RP9 — Light (Reflection and Refraction)

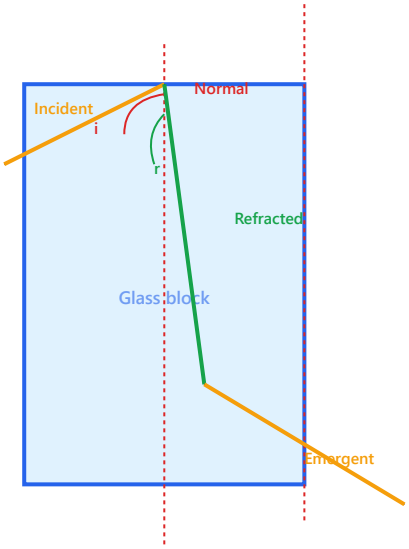
> **Aim:** Investigate the law of reflection (angle of incidence = angle of reflection) and refraction through a glass block (angle changes when light passes between media of different optical densities).

Apparatus

- Ray box or laser pointer (low-power, e.g. <5 mW)
- Rectangular glass block (or semicircular glass block for refraction)
- Protractor (for measuring angles)
- Plain white paper
- Pencil (sharp, for drawing rays)
- Mirror (plane, flat) for reflection experiments

- Ruler
- Dark room or screen (to improve ray visibility)

RP9: Light — Refraction Diagram



Key Observations:

Refraction:

- Entering glass (denser medium): ray bends toward the normal
- Leaving glass (less dense): ray bends away from normal
- Angle inside glass (r) is smaller than angle in air (i)
- Emergent ray parallel to incident ray (laterally displaced)

Snell's Law:

$n_1 \sin i = n_2 \sin r$

(where n = refractive index)

Method — Reflection

1. Place a plane mirror on a sheet of white paper.
2. Draw a line perpendicular to the mirror surface (the **normal**).
3. Using the ray box, shine a light ray at the mirror at a shallow angle (e.g. 30° from the normal).
4. Trace the incident ray and the reflected ray on the paper.
5. Using a protractor, measure the **angle of incidence** (i) and the **angle of reflection** (r) from the normal.
6. Verify: **angle of incidence = angle of reflection** (law of reflection)
7. Repeat at different angles (e.g. 20°, 40°, 50°, 60°).

Method — Refraction

1. Place a rectangular glass block on a sheet of white paper and draw around its outline with a pencil.

2. Remove the glass block. Draw a normal line perpendicular to one side of the outline (the point of incidence).
3. Place the glass block back on the outline.
4. Using the ray box, shine a light ray toward the glass block at an angle (e.g. 40° from the normal).
5. Look at the emergent ray on the opposite side and trace it on the paper.
6. Remove the glass block and draw:
   - The incident ray (from ray box to block)
   - The refracted ray (inside the block, based on the light path you observed)
   - The emergent ray (from block exit to where it hits the paper)
7. Measure:
   - **Angle of incidence (i):** between incident ray and normal
   - **Angle of refraction (r):** between refracted ray and normal inside the block
8. Verify: **r < i** (ray bends toward normal entering denser medium)
9. Calculate the **refractive index**: `n = sin i / sin r` (for glass, typically ~1.5)
10. Note: The emergent ray is parallel to the incident ray but laterally displaced.
11. Repeat at different angles (e.g. 30°, 50°, 60°) to gather more data.

Key Measurements and Calculations

- **Angle of incidence (i):** angle between incident ray and normal, in degrees
- **Angle of reflection (r):** angle between reflected ray and normal, in degrees
- **Law of reflection:** i = r (for reflection)
- **Angle of refraction (θ):** angle between refracted ray and normal inside the block, in degrees
- **Snell's Law:** $n_1 \sin i = n_2 \sin \theta$ (where $n_1 = 1.0$ for air, $n_2 \approx 1.5$ for glass)
- **Refractive index of glass:** $n = \sin i / \sin \theta \approx 1.5$ (typical value)

Sources of Error & Minimisation

- **Accuracy of angle measurement:** Use a sharp pencil and a clear protractor. Ensure the normal is drawn perpendicular to the surface (use a set square or ruler).
- **Ray box alignment:** Ensure the ray is narrow and clearly defined. Use a slit if necessary.

- **Light scattering in glass:** Minor light scattering can make the ray path less distinct. Use a dark room and keep the glass block clean.
- **Lateral displacement:** The emergent ray's position depends on the glass block thickness. Keep thickness constant when repeating measurements.

Safety Considerations

- **Laser safety:** If using a laser pointer, avoid looking directly into the beam. Do not shine it at reflective surfaces where it could reflect into eyes.
- **Glass block:** Glass blocks are fragile; handle carefully to avoid breakage and sharp edges.

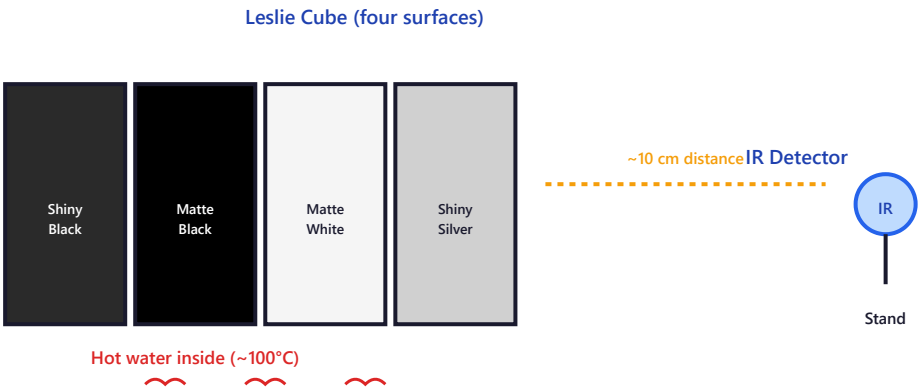
RP10 — Infrared Radiation (Leslie Cube)

> **Aim:** Investigate how the rate of infrared radiation emission depends on the surface properties (color and texture). Demonstrate that matte black surfaces are the best emitters of infrared radiation; shiny surfaces are poor emitters.

Apparatus

- Leslie cube (a hollow metal cube with four different surfaces):
  - Shiny black (matt black paint)
  - Matte black (dull black paint)
  - Shiny silver (polished or silvered surface)
  - Matte white (white paint)
- Boiling water (or hot water from a kettle or Bunsen burner)
- Infrared detector (thermopile or IR sensor) or infrared thermometer
- Data logger with temperature sensor (optional)
- Ruler (for measuring distance from cube to detector)
- Thermometer (to confirm water temperature)
- Stand or support for IR detector (to position at fixed distance)

RP10: Infrared Radiation (Leslie Cube) — Apparatus



Procedure:

1. Fill Leslie cube with boiling water (~100°C). 2. Position IR detector at fixed distance (~10 cm) perpendicular to each face. 3. Record IR reading for each su

4. Compare readings: Matte black > Matte white > Shiny surfaces. 5. Repeat as cube cools to show relative emission rates remain constant.

Conclusions: Matte black = best emitter (absorber) of IR. Shiny/light surfaces = poor emitters (reflectors). Dark, rough surfaces absorb and emit more IR th

Method

1. Fill the Leslie cube with boiling water (approximately 100°C). Allow water to heat all four surfaces uniformly for a minute.
2. Check that the temperature inside is stable using a thermometer inserted into the cube.
3. Position the infrared detector (thermopile or IR sensor) at a fixed distance from the cube (e.g. 10 cm perpendicular to the surface). Use a ruler to ensure consistency.
4. Point the detector at the **shiny black surface** and record the IR reading (or data logger output) after waiting ~10–15 seconds for stabilization. Record the reading.
5. Reposition the detector to face the **matte black surface** (at the same distance) and record the reading.
6. Repeat for the **matte white surface** and **shiny silver surface**.
7. Tabulate the results for each surface at the same water temperature.
8. Repeat the measurements as the cube cools (e.g. every 5 minutes for 20 minutes) to observe how emission rates change with temperature. The relative order should remain the same (matte black highest, shiny lowest).
9. Plot a graph of IR reading vs. surface type (bar chart) and vs. temperature (line graph) for each surface.

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Key Measurements and Calculations

- **Temperature of cube:** measured with thermometer, in °C or K
- **IR detector reading:** voltage (mV) or relative units proportional to radiated power
- **Surface comparison:** Relative IR readings show which surfaces emit most:
  - Matte black: highest reading (best emitter)
  - Matte white: medium reading
  - Shiny black: lower reading (some reflection)
  - Shiny silver: lowest reading (poor emitter, good reflector)
- **Temperature effect:** As cube cools, all IR readings decrease, but relative order remains the same.

Sources of Error & Minimisation

- **Detector distance:** Always position the detector at the same distance from each surface (~10 cm). Measure with a ruler to ensure consistency. IR intensity decreases with the square of distance (inverse square law).
- **Detector alignment:** Point the detector perpendicular to each surface; angled approaches reduce the effective radiation received.
- **Temperature variation:** Measure the water temperature before and after each set of readings. Ideally, take all four readings in quick succession to minimize cooling.
- **Ambient IR radiation:** Reflections from the environment (other hot objects, walls, sun) can affect readings. Conduct experiment in a thermally controlled room if possible, and shield detector from other heat sources.
- **Detector calibration:** Check that the IR detector is working correctly by comparing readings with a known heat source.

Safety Considerations

- **Boiling water:** Handle with care; use heat-resistant gloves or tongs. Avoid splashing.
- **Hot cube:** The Leslie cube becomes very hot; do not touch directly. Allow to cool before handling.
- **Water spillage:** Keep the work area dry to prevent slipping.

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